

Applied NLP

DATA 641

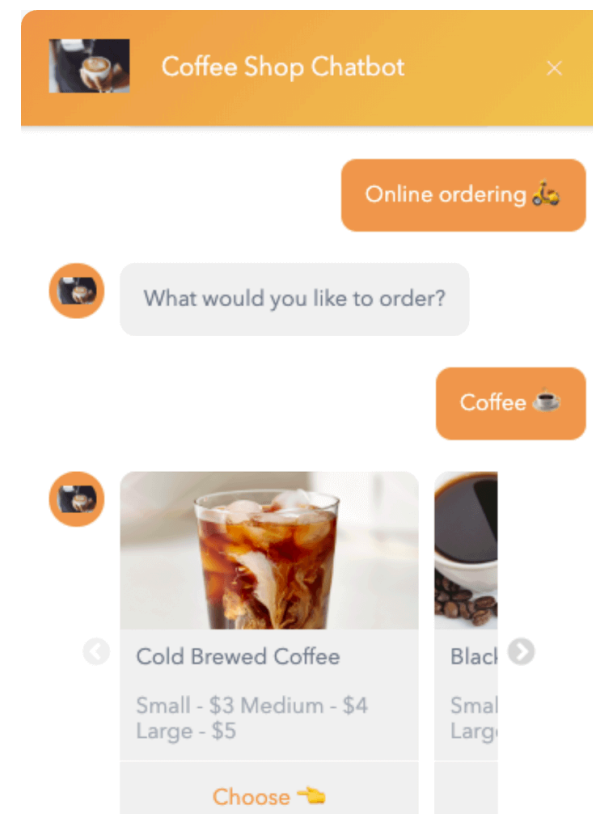
Lecture 10 — Information Extraction

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

Information Extraction (IE) applications

IE is used in a wide range of real-world applications

- **Tagging news and other content**
- **Chatbots**
 - A chatbot needs to understand the user's question in order to generate/retrieve a correct response. For example, consider "Online ordering" The chatbot needs to understand that we need to order online and generate the question "What would you like to order?"



- **Applications in social media**

- A lot of information is disseminated through social media channels. Extracting informative excerpts from social media text may help in decision making. An example use case is extracting time-sensitive, frequently updated information, such as traffic updates and disaster relief efforts, based on tweets.



- **Extracting data from forms and receipts**

- Many banking apps nowadays have the feature to scan a check and deposit the money directly into the user's account. Whether you're an individual, small business.

IE tasks

IE is a term that is used to refer to a range of different tasks of varying complexity

For a human reader, the following text talks about

- The company Apple (not the fruit)
- Luca Maestri(finance chief)
- Buyback of stock

SAN FRANCISCO — Shortly after Apple used a new tax law last year to bring back most of the \$252 billion it had held abroad, the company said it would buy back \$100 billion of its stock.

Rectangular Snip

On Tuesday, Apple announced its plans for another major chunk of the money: It will buy back a further \$75 billion in stock.

“Our first priority is always looking after the business and making sure we continue to grow and invest,” Luca Maestri, Apple’s finance chief, said in an interview. “If there is excess cash, then obviously we want to return it to investors.”

Apple’s record buybacks should be welcome news to shareholders, as the stock price is likely to climb. But the buybacks could also expose the company to more criticism that the tax cuts it received have mostly benefited investors and executives.

For a machine to understand all this involves different levels of IE

- Identifying that article is about "buyback" or "stock price" relates to keyword or key phrase extraction (KPE)
- Identifying Apple as an organization and Luca Maestri as a person relates to named entity recognition (NER)
- Recognizing Apple as Apple, Inc and not another company with the name Apple refers to named entity disambiguation and linking
- Extracting information that Luca Maestri is the finance chief is relation extraction
- Identifying that article is about a single event refers to event extraction
- Another task is temporal information extraction which aims to extract info about times and dates and is useful for developing calendars